

Supplementary Material: Probing-Based Test-Time Steering of Music Diffusion Transformers

Junyoung Koh^{1 2 3}

1. Experimental setup

Table 1 summarizes the full configuration shared by all experiments in the main text. A single corpus and a single set of intermediate activations from a frozen Stable Audio Open DiT are reused across probing, sparse-autoencoder training, and steering. Each prompt–concept sweep uses a single fixed seed, held constant across steering strengths so that differences are attributable to the intervention. Extending to multi-seed estimates is left to future work.

2. Sparse-autoencoder baseline

The main text builds on probe-based directions. As a complementary baseline we applied the now-standard sparse-autoencoder (SAE) recipe to the same activations. Here we report why it does not yield usable concept handles under our budget.

Method. For each studied layer we train a TopK SAE (Gao et al., 2024) on the same activations used for probing. The SAE reconstructs an activation h as

$$\hat{h} = W_d \text{TopK}_k(W_e(h - b_d)) + b_d, \quad (1)$$

with 64 active features per token and an $8\times$ overcomplete dictionary of 12,288 features. An AuxK auxiliary loss on the top 256 dead features, weighted by $1/32$, prevents feature collapse, and we optimize with Adam (Kingma & Ba, 2015) at learning rate 3×10^{-4} for 100 epochs. To build a steering direction for concept k , we score each feature j by its positive-versus-negative activation contrast,

$$\Delta_j(k) = \bar{a}_j^+(k) - \bar{a}_j^-(k), \quad (2)$$

where $\bar{a}_j^\pm(k)$ is the mean activation of feature j on time-pooled tokens of clips labeled positive or negative for k . We take the top five features by $\Delta_j(k)$ and use the ℓ_2 -normalized sum of their decoder columns as the steering

¹Department of Artificial Intelligence, Yonsei University, Republic of Korea ²MAAP Lab ³KRAFTON. Correspondence to: Junyoung Koh <solbon1212@yonsei.ac.kr>.

Table 1. Experiment setup. All components share a single corpus and a single set of intermediate activations from a frozen Stable Audio Open DiT.

Data	
Corpus	150 clips, 10 s each
Sources	100 MusicCaps + 50 MUSDB18-HQ
Sample rate	44.1 kHz
Model (Stable Audio Open, frozen)	
DiT depth L	24 transformer blocks
Hidden dim D	1536
VAE downsample	$1920\times$
Studied layers ℓ	$\{6, 12, 18\}$
Snapshot timestep t^*	0.5
Concept probing	
Concepts	13 keyword lists, 11 probed
Labels	caption keyword matching
Classifier	class-balanced logistic regression
Evaluation	5-fold stratified CV (AUC)
Sparse autoencoder	
Architecture	TopK (Gao et al., 2024) + AuxK
Active features k	64
Dictionary size d_{sae}	12,288 ($8\times$ overcomplete)
Training	Adam, lr 3×10^{-4} , 100 epochs
Steering experiments	
Hook layer	$\ell \in \{6, 12, 18\}$
Strength α	main sweep $\{-30, -15, 0, 15, 30\}$; CLAP at 10, 15, 30
Steered concepts	drums, guitar, bass, electronic, vocal
Base prompts	1 neutral + 5 musical genres
Sampling	100 diffusion steps, cfg 7.0

direction, holding hook, layer, and strength identical to the probe-based protocol. The goal is not to claim that SAEs cannot work for music DiTs at larger scale, but to test whether the standard small-to-medium activation-analysis recipe surfaces usable concept handles under the same budget in which linear probes already do.

Result. The SAEs train cleanly, with moderate reconstruction and no dead features (Table 2), yet their steering directions are far weaker than the probe-based ones. On the same five concepts they yield slopes of only 1.9 to 5.9, roughly 3.6 to 8.5 times smaller than the probe directions, with response curves that are non-monotonic or flat (Figure 1). Under this budget the SAE does not surface the clean concept handles

that linear probes already provide.

Diagnostics. The diagnostics in Table 2 point to two shortcomings. We define temporal sharpness as a feature’s peak frame activation divided by its mean activation over frames where it fires, so a value near one indicates diffuse firing across a clip rather than a sharply localized event. First, the features are not temporally localized: with a TopK constraint each feature is active on only about 0.5% of frames, the fraction of the dictionary left active, but the near-unity median sharpness shows that an active feature spreads almost uniformly across a clip rather than marking a localized event such as a drum hit, so its decoder column is a poor template for an instrument that is intermittent in time. Second, no feature is concept-selective: the best top-1 versus top-2 concept margin stays below five at every layer, meaning even the most concept-aligned feature responds nearly as strongly to a competing concept, so summing the top five decoder columns blends several concepts into one direction and dilutes the intended one. Reconstruction error also grows with depth, from a fraction of variance unexplained of 0.16 at layer 6 to 0.32 at layer 18, so the dictionary captures progressively less of the signal in the very layers where probing stays accurate. Taken together, the SAE distributes each musical concept across many weakly selective, temporally diffuse features rather than isolating it in a few monosemantic ones, which is exactly the structure that linear probes exploit and that additive SAE-feature steering cannot. We attribute the gap to music activations being denser than the per-token text activations on which SAEs were validated, to the TopK parametrization being ill-suited to audio at this scale (gated alternatives such as JumpReLU (Rajamanoharan et al., 2024) may fit better), and to our single-timestep collection at the midpoint t^* exposing a narrower representational range than the full diffusion trajectory.

Table 2. Sparse-autoencoder summary at three layers. Top block: training metrics. Bottom block: feature diagnostics. SAEs train successfully but features are neither temporally localized nor concept-selective, and yield much weaker steering than probe directions.

	L6	L12	L18
Val FVU	0.16	0.24	0.32
Dead features	0/12K	0/12K	0/12K
Mean firing rate	0.5%	0.5%	0.5%
Sharpness median	1.08	1.10	1.13
Concept margin max	2.60	1.87	3.28
SAE steering slope (mean)	—	+3.9	—
Probe steering slope (mean)	+24.0	+18.2	+17.5

3. Cross-layer steering

Table 3 gives the per-concept steering slopes at the three injection layers summarized in the main text. Earlier injection

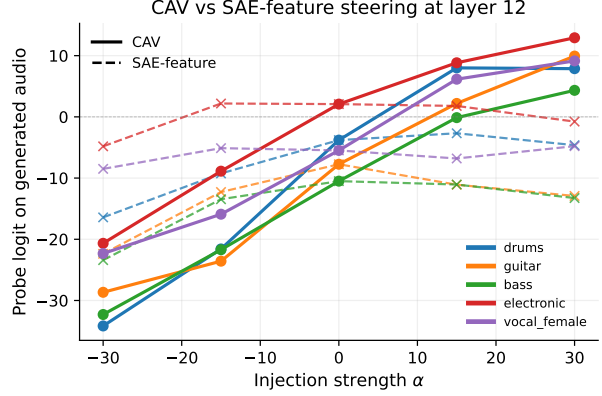


Figure 1. Probe-score response curves at layer 12 for the five tested concepts under CAV steering (solid) and SAE-feature steering (dashed). CAV curves are monotonic with α and have large dynamic range, whereas the matched SAE-feature directions are nearly flat under the same activation budget.

yields larger effects.

Table 3. Mean slope per $\pm 30\alpha$ across five concepts, by injection layer. Earlier injection yields larger effects.

ℓ	drums	guitar	bass	electronic	vocal_f	mean
6	17.7	26.2	27.1	25.3	23.8	24.0
12	21.0	19.3	18.3	16.8	15.7	18.2
18	20.8	21.8	16.7	11.7	16.7	17.5

Per-layer probe AUC. The steering experiments use the three layers $\ell \in \{6, 12, 18\}$. To verify that this is not an arbitrary choice, we also collect activations at every one of the 24 DiT blocks on the same 150-clip corpus and fit a probe at each (Figure 2). Mean AUC across the eleven concepts rises from about 0.72 at the first layer to a peak of 0.77 around layer 5, then settles into a plateau in the range 0.74–0.75 from layer 7 onward. Individual concepts show the same shape: instruments such as drums, bass, and electronic peak in the first quarter of the network and then drop a few points, while structural categories such as classical stay nearly flat. Our chosen $\ell = 6$ sits at the high end of this curve, layer 12 is representative of the mid-network plateau, and layer 18 is a late but still informative layer, so the three studied layers cover the range of probe quality the DiT exhibits.

4. Cross-prompt steering

Table 4 gives the full per-prompt Δ probe scores summarized in the main text. All fifteen prompt-concept pairs are positive, with moderate within-concept variance.

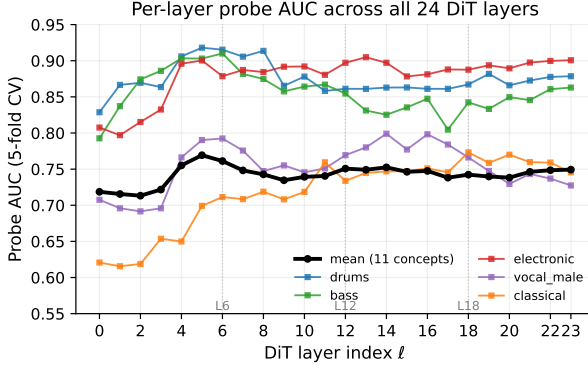


Figure 2. Probe AUC at every DiT layer (mean across eleven concepts, plus five representative concepts). The vertical dotted lines mark the layers $\ell \in \{6, 12, 18\}$ used in the steering experiments.

Table 4. Δ probe score, defined as the score at steering strength $+30$ minus the score at baseline, on five base prompts (layer 12). All entries positive with moderate within-concept variance.

Prompt	bass	drums	electronic
jazz trio	15.2	16.3	16.7
energetic rock	14.5	10.6	21.4
classical orchestra	14.1	9.5	15.6
ambient soundscape	16.7	18.3	15.2
hip-hop beat	19.5	5.6	11.2
mean $\pm$ std	16.0 $\pm$ 2.0	12.1 $\pm$ 4.6	16.0 $\pm$ 3.3

5. Timestep-windowed injection and distributional drift

Our default protocol injects $\alpha \tilde{v}_k$ at every one of the 100 denoising steps. With large α this accumulated perturbation pushes activations off the data distribution and the audio degrades, which is the over-steering we report in the main text at $\alpha = 30$. A simple variant restricts the injection to a sub-window of denoising steps, for example only steps 30 through 70. The model can then settle back to its in-distribution trajectory during the unperturbed final steps. Table 5 reports the centroid distance between CLAP embeddings of baseline and steered audio, averaged across our five cross-prompt base prompts and three concepts. Plain steering at strength 30 doubles the centroid distance relative to strength 15, but windowed steering at strength 25 keeps the centroid distance close to the baseline-strength setting while still producing larger per-concept CLAP changes. The picture is consistent on the per-concept side: at steps 30–70 and strength 25, the electronic addition direction grows from $\Delta\text{CLAP} = +0.001$ at strength 15 to $+0.040$, and the electronic suppression direction reverses from a CLAP-detected failure at strength -15 to a clean -0.031 at strength -25 , all without any increase in distributional drift.

As a complementary distributional check that does not rely

Table 5. Distributional check between baseline and steered audio, averaged across the same five base prompts and three concepts. Centroid distance is the ℓ_2 distance between CLAP embedding centroids; MAUVE (Pillutla et al., 2021) is adapted to CLAP audio features and higher values mean the steered distribution stays closer to the baseline. The windowed variant at strength 25 keeps both metrics close to the moderate-strength setting, while the full-trajectory $\alpha = 30$ run drifts on the centroid metric and collapses the MAUVE score.

Setting	Centroid L2	MAUVE
$\alpha = -15$, full trajectory	0.188	1.000
$\alpha = +10$, full trajectory	0.225	1.000
$\alpha = +15$, full trajectory	0.259	1.000
$\alpha = -25$, window [30, 70]	0.296	1.000
$\alpha = +25$, window [30, 70]	0.316	1.000
$\alpha = +30$, full trajectory	0.511	0.329

on a single CLAP cosine, we adapt MAUVE (Pillutla et al., 2021) to CLAP audio features, computed between the set of steered clips and the corresponding baseline clips. The right column of Table 5 reports the mean MAUVE across the three concepts. With our small per-condition sample the metric saturates near one for all moderate and windowed settings, but the strongest non-windowed setting at $\alpha = 30$ drops to 0.33, mirroring the rise in centroid distance and suggesting that the windowed protocol stays closer to the natural baseline distribution than the strongest non-windowed setting does.

6. Prompt richness vs steering effect

The quality–steerability tradeoff reported in the main text and in Section 12 comes out as a clean monotone curve when we tabulate three matched settings on the same three trials (electronic on funk, drums on classical, electronic on rock). Table 6 pools the CLAP signal, the single-listener accuracy on the same listener, and the listener’s baseline audio-quality rating. The pattern is consistent. Going from short prompts at 100 denoising steps to rich prompts at 200 denoising steps gradually doubles the listener’s baseline quality rating, the CLAP magnitude shrinks by an order of magnitude, and the listener accuracy drops in this small pilot. The intermediate setting of rich prompts at 100 steps gives most of the quality improvement and still preserves a perceptible if smaller steering signal, and is the configuration our pilot uses for the final listening trials.

7. Classifier-free guidance ablation

The relative strength of text conditioning and probe-based steering is controlled in part by the classifier-free guidance scale (Ho & Salimans, 2022). We sweep cfg in $\{3, 5, 7, 10\}$ on the same bass-on-jazz cross-prompt steering at $\alpha = +15$ and report the CLAP change in Table 7. The result is an

Table 6. Quality vs steerability across three matched settings on the same three trials. Higher MOS means better baseline audio quality. “Accuracy” is the single listener’s pairwise accuracy on the three trials. Mean $|\Delta\text{CLAP}|$ takes the absolute CLAP change before averaging so a reversed CLAP signal still registers as a sizeable magnitude.

Setting	Mean $ \Delta\text{CLAP} $	Listener accuracy	Baseline MOS
short prompts, 100 steps, $\alpha=25$ window	0.12	3/3	moderate
rich prompts, 100 steps, $\alpha=22$ window	0.04	2/3	high (funk 4/5)
rich prompts, 200 steps, $\alpha=22$ window	0.04	4/10	high

inverted-U pattern. Our default cfg of 7 matches Stable Audio Open’s recommended sampling regime and gives a clear positive $\Delta\text{CLAP} = +0.095$, with a slightly larger $+0.138$ at cfg of 5. Below this, at cfg 3, the text condition is too weak and the baseline audio drifts into a regime where the bass direction no longer maps onto a clean positive change. Above it, at cfg 10, the text condition saturates the audio so strongly that the same additive steering pushes the score in the opposite direction. The takeaway is that the dominance of the text condition is itself a knob: cfg and steering strength can be tuned together as a complementary control axis, with the default cfg sitting at a robust point inside the productive range rather than at its peak.

Table 7. CFG sweep for bass on a smooth jazz cross-prompt at $\alpha = +15$. Both very weak (cfg 3) and very strong (cfg 10) text conditioning reverse the steering, while cfg 5 to 7 gives the largest positive change.

cfg	baseline	steered	Δ
3	0.305	0.282	-0.023
5	0.189	0.327	+0.138
7 (default)	0.143	0.238	+0.095
10	0.236	0.135	-0.101

8. Multi-layer simultaneous injection

A natural follow-up is whether injecting at several DiT blocks at once linearly amplifies the steering, or whether a single layer already exhausts the controllable range. We compare three protocols on the same $\alpha = +15$ cross-prompt sweep used in the main text: injection at $\ell = 6$ alone, at $\ell \in \{6, 12\}$ simultaneously, and at $\ell \in \{6, 12, 18\}$ simultaneously. Table 8 reports mean ΔCLAP across three base prompts and three target concepts.

Stacking layers does not give the additive amplification one might naively expect. The mean across the nine prompt-concept pairs actually moves further from the baseline distribution as more blocks are hooked in, and the within-pair pattern is unchanged: trials where the text condition leaves headroom remain positive while trials where the text explicitly evokes the targeted concept (drums and bass on the hip-hop prompt) remain negative. This is consistent with the cumulative drift picture from Section 5: more layers injected

Table 8. Mean ΔCLAP across nine prompt-concept pairs when the same $\alpha = +15$ injection is applied at one, two, or three DiT blocks simultaneously. Stacking more layers does not increase the steering signal and the mean drifts further from baseline, consistent with the cumulative off-distribution drift we observe at large single-layer α .

Injection layers	Mean ΔCLAP
$\ell = 6$	-0.020
$\ell \in \{6, 12\}$	-0.036
$\ell \in \{6, 12, 18\}$	-0.075

at once is, in effect, a stronger perturbation that pushes the activations further off the natural text-conditioned trajectory, fighting the same cross-attention that an early single-layer injection does not directly confront.

9. Steered spectrograms

Figure 3 shows mel-spectrograms of steered generations, illustrating the time–frequency changes that accompany the probe-score and CLAP changes reported in the main text.

10. Concept keyword lists

Table 9 lists the caption keywords used for weak labeling of each concept. A clip is labeled positive for a concept if its caption contains any of the listed substrings (case-insensitive). Concepts with fewer than five positive clips in the 150-clip corpus are excluded from the probing and steering experiments.

Table 9. Caption-keyword lists used to weakly label the 150-clip analysis corpus.

Concept	Keywords (case-insensitive substring match)
jazz	jazz, saxophone, blues
rock	rock, metal, punk
electronic	electronic, edm, synth, techno, hip-hop, hiphop, trap
classical	classical, orchestra, piano sonata, string
folk	folk, acoustic, country, guitar fingerpick
ambient	ambient, drone, soundscape
vocal_male	male voice, male vocal, male sing
vocal_female	female voice, female vocal, female sing
drums	drum, percussion, kick, snare
bass	bass
guitar	guitar
piano	piano, keyboard
strings	string, violin, cello

11. CLAP validation details

The independent acoustic check uses the frozen `laion/clap-htsat-unfused` checkpoint (Wu et al., 2023). Generated audio is resampled to 48 kHz and

Mel-spectrograms of steered generations (layer 12)

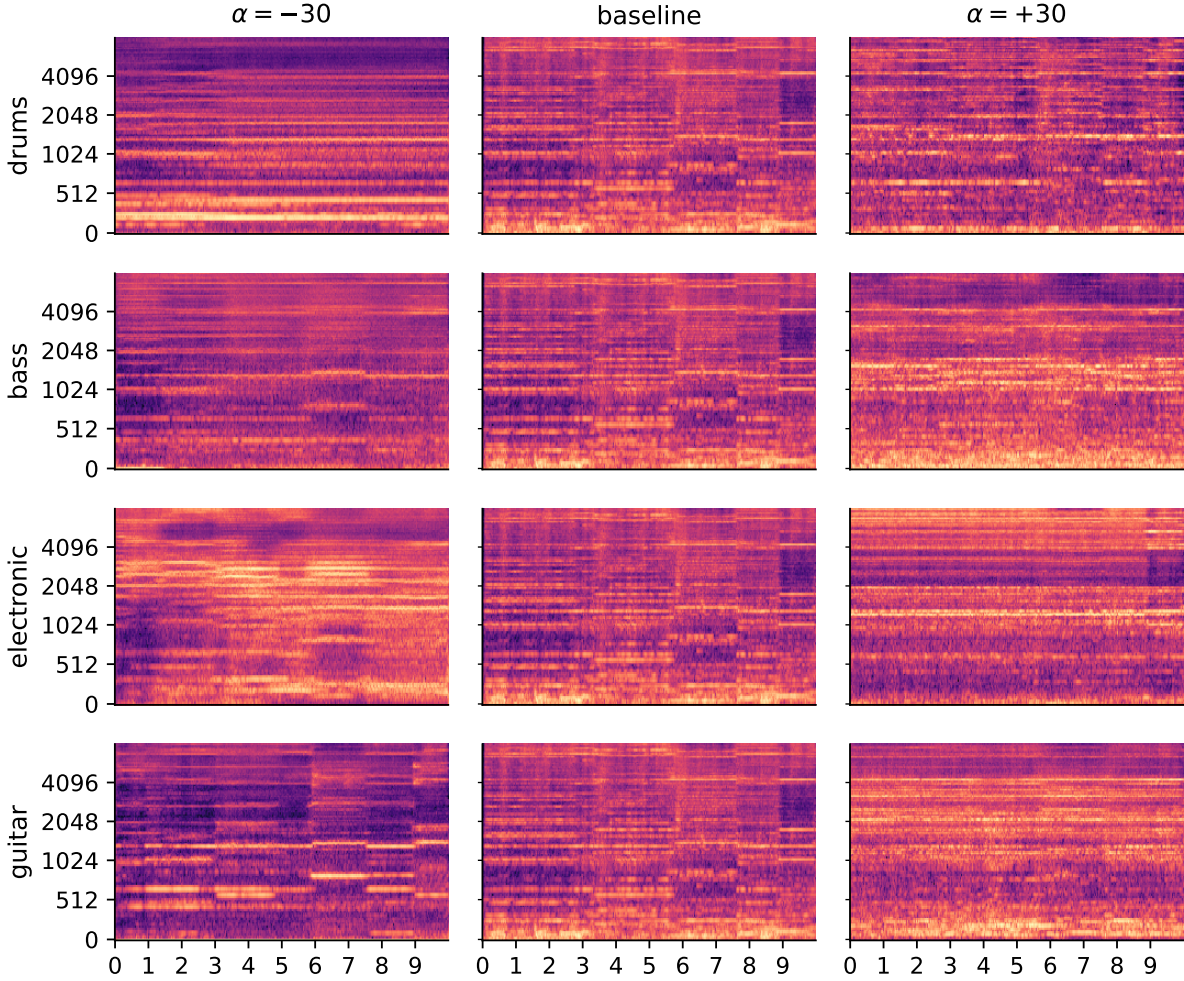


Figure 3. Mel-spectrograms of generations under CAV steering at layer 12. Rows are concepts and columns are negative, zero, and positive steering. Relative to the center column, negative steering thins out the targeted content, while positive steering adds concept-related energy such as stronger drum transients, low-frequency bass, denser electronic high-frequency content, and additional guitar harmonics. At the largest positive strength this also begins to smear the spectrogram, consistent with the CLAP decline reported in the main text.

embedded with the CLAP audio encoder. The concept text prompts are: “music with prominent drums and percussion” (drums), “music with a strong bassline and low-frequency bass” (bass), “electronic music with synthesizers” (electronic), “music featuring electric guitar” (guitar), and “music with female vocals singing” (vocal.female). We report cosine similarity between the ℓ_2 -normalized audio and text embeddings.

Table 10 reports the single-prompt α sweep. Here the baseline already attains moderate-to-high concept similarity (for example 0.49 for electronic), because the neutral prompt “instrumental music” tends to generate audio that already contains these common elements. With little headroom to add, positive steering does not raise the similarity, whereas

negative steering drives it toward zero, so this single-prompt setting most clearly demonstrates suppression rather than addition. To measure addition we therefore use the cross-prompt setting, where the base prompt leaves room for the concept. Table 11 reports that analysis at three positive strengths, averaged over the five base prompts. At moderate strength, CLAP similarity increases for some concepts and prompt sets, most clearly bass and electronic at a strength of 10, while drums remains mixed. At the strongest setting of 30 the change turns negative for all three concepts, consistent with strong positive injection moving generations off-distribution rather than adding clean concept content.

Table 10. CLAP concept-text similarity of single-prompt steered audio at layer 12, across injection strength α .

Concept	Steering strength				
	−30	−15	0	+15	+30
drums	−0.04	0.19	0.24	0.18	0.15
bass	−0.02	0.12	0.20	0.33	0.11
electronic	0.12	0.46	0.49	0.31	0.17
guitar	0.01	0.15	0.24	0.20	0.13
vocal_female	0.13	0.27	0.26	0.20	0.13

Table 11. Cross-prompt CLAP change from baseline to positive α , averaged over the five base prompts (mean $\pm$ standard deviation; in parentheses, number of prompts with a positive change). Moderate steering at strength 10 increases CLAP similarity for electronic and bass, while the strongest steering at 30 is net negative, consistent with off-distribution degradation.

Concept	Steering strength		
	+10	+15	+30
drums	−0.004 (2/5)	+0.010 (3/5)	−0.010 (2/5)
bass	+0.060 (3/5)	+0.022 (3/5)	−0.046 (1/5)
electronic	+0.032 (5/5)	+0.001 (2/5)	−0.004 (2/5)

12. Single-listener listening pilot

We use this only as a qualitative sanity check, not as evidence for the size or sign of the perceptual effect. The experiment is a single-listener pairwise pilot with the author as the listener, built from the cross-prompt generations at moderate strength. We do not run a statistical test, do not estimate effect sizes, and we keep the protocol in the supplementary material rather than the main text so it is not mistaken for a controlled perceptual study. A full multi-rater listening study is required to draw any quantitative claim about audibility, and is listed in the main text as the most important next step. With those caveats in place, we describe what one listener heard.

Each trial pairs a baseline clip with a steered clip on a musical base prompt (jazz, rock, classical, ambient, or hip-hop) and asks which clip has more, less, or about the same amount of a target concept. The single listener identified the steered clip on easy, musically natural tasks, namely removing an instrument that is prominent in the genre (drums or bass from rock and hip-hop, electronic from ambient and hip-hop) or strengthening one that is already plausible (bass in jazz and ambient), and was close to chance on tasks that require synthesizing an instrument the genre normally lacks, such as adding drums to a classical orchestra. This qualitative split mirrors the CLAP analysis: the effect is more audible when the audio has headroom for the concept to change, and weaker when steering must create content from nothing. Audio-quality ratings were moderate, between two and four on a five-point scale. The quality–steerability

tradeoff observed under richer prompts and longer denoising trajectories is reported separately in Section 6.

13. Cross-model probe on ACE-Step turbo

The limitations note in the main text flags that we study one DiT (Stable Audio Open 1.0) and that generalization to other modern music DiTs remains to be checked. As a first cross-model probe we repeat the activation-collection, probing, and CAV-steering protocol on ACE-Step v1.5 turbo, an open music diffusion transformer with $L=24$ blocks and hidden dimension $D=2048$. The corpus, caption-keyword labels, snapshot timestep $t^*=0.5$, and probing-classifier configuration are identical to the SAO experiments; only the model, its VAE pretransform, and its text/lyric encoder differ. We report this as a preliminary single-checkpoint follow-up, not a controlled multi-model study.

Probes remain informative but layer dependence flattens.

The per-layer probe AUC on ACE-Step, averaged across the same eleven concepts, sits in a narrow band of 0.686 to 0.705, with a slight U-shape: layer 0 gives 0.705, the mid-network plateau is around 0.69, and the late layers $L20$ – $L23$ recover to 0.702–0.705. Two observations follow. First, all 24 layers are usefully above chance on the same caption-keyword task, so concept information is broadly present in ACE-Step’s residual stream just as in SAO. Second, the early-layer peak we report for SAO (mean AUC near 0.77 around layer 5, falling to 0.74–0.75 on the plateau) does not appear on ACE-Step. The depth-strength relationship is flatter and lower in absolute terms. A plausible contributing factor is ACE-Step’s AdaLN scale-shift modulation conditioned on the timestep embedding, which re-projects the residual stream at every block and may homogenize the per-layer geometry that probes exploit.

Causal CAV steering is far less specific. Treating the same probe coefficients as steering directions and injecting at layer 12 with $\alpha \in \{-30, -15, 0, +15, +30\}$ (matching the main-text protocol), the monotonic concept-presence response we observe on SAO is weaker on ACE-Step and is sometimes non-monotonic (for example the vocal-female direction reaches its maximum probe-score change near $\alpha = \pm 15$ rather than at the extremes). More telling is the specificity matrix at $\alpha = +30$, summarized in Table 12. The diagonal mean Δ probe score is +0.34 and the off-diagonal mean $|\Delta|$ is 0.35, giving a diagonal-to-off-diagonal ratio of approximately 0.95. The matched SAO setting yields a ratio of 3.6 (main-text Table 3), so on this single ACE-Step checkpoint the steering directions are essentially as likely to perturb a competing concept probe as their own. Several rows even show their largest absolute response on a non-target concept: the drum and electronic steering directions both produce their largest measurement on the guitar

probe, and the electronic direction in fact registers as a small *negative* change on the electronic probe at $\alpha = +30$. The polarity of the steering also matters more on ACE-Step than on SAO: the suppression direction $\alpha = -30$, which the main text reports as the cleanest of the SAO operations, is the most broken on ACE-Step, with only two of the five concept probes responding in the intended negative direction. Whatever residual specificity remains at this layer is concentrated on the positive-strength side.

Table 12. Δ probe score on ACE-Step at $\alpha = +30$, layer 12, for five concept steering directions (rows) measured on the same five concept probes (columns). Diagonal entries that should dominate are shown with a $\bullet$ prefix. The mean diagonal change is +0.34 and the mean off-diagonal magnitude is 0.35, a ratio of 0.95 versus 3.6 on SAO.

steer \ meas	drums	bass	electronic	guitar	vocal_f
drums	$\bullet +1.07$	+0.47	-0.56	+1.36	-0.42
bass	+0.40	$\bullet -0.47$	-0.44	+0.63	-0.24
electronic	+0.75	+0.68	$\bullet -0.29$	+1.25	-0.03
guitar	+0.71	-0.50	-0.45	$\bullet +1.28$	-0.09
vocal_f	+0.24	+0.01	+0.01	+0.48	$\bullet +0.08$

The specificity collapse is consistent across depth. We repeat the steering matrix at layers 6 and 18 with the same five concepts and the same $\alpha = +30$ readout. Table 13 reports the layer-wise diagonal mean, off-diagonal mean magnitude, and their ratio. At layer 6 the diagonal mean is in fact *negative*, meaning the average effect of steering a concept on its own probe at this depth is in the opposite of the intended direction, and the ratio is -0.84 . The middle layer 12 gives a ratio of 0.95, and the late layer 18 recovers to 0.59. Even the most favourable layer remains well below the SAO ratio of 3.6 at every block we tested; the diagonal-dominance failure is not a quirk of one injection depth.

Table 13. Layer-wise specificity on ACE-Step at $\alpha = +30$. Diagonal mean is the average Δ probe score of each steering direction on its own concept probe; the off-diagonal mean magnitude is the average over all non-target cells. Even the most favourable layer ($\ell = 18$, ratio 0.59) is well below the SAO ratio of 3.6 reported in Table 3 of the main text, and the $\ell = 6$ ratio is in fact negative.

Layer	diag mean Δ	off-diag mean $ \Delta $	ratio
$\ell = 6$	-0.341	0.406	-0.84
$\ell = 12$	+0.335	0.354	+0.95
$\ell = 18$	+0.201	0.340	+0.59

Interpretation. The combination of broadly-present-but-flatter probes, weaker-and-much-less-specific causal steering, and the polarity asymmetry above is consistent with a model whose internal concept geometry is reshaped at each block by AdaLN modulation. Linear probes can still locate concept signal, but the additive residual-stream intervention that works cleanly on SAO does not pick out a single concept axis on ACE-Step at any of the three layers we tested, and

the negative-strength side of the protocol is the first to fail. We report this as an early cross-model observation rather than a definitive claim: a single checkpoint, a single neutral prompt, and five concepts are all clear limitations. Future work could explore variants of the hook that are aware of the model’s own routing of conditioning information (for example a per-frame gate derived from the live activation, or interventions that target the cross-attention path rather than the residual stream), and extend the cross-model panel to other modern music DiTs.

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